Logan Jaeger
New York City, NY, USA
650-556-4570 | loganjaeger@gmail.com

Education

University of California, Berkeley

B.A., Physics; 2017-2021 — Graduate with Distinction in General Scholarship

Columbia University

M.A., Statistics; 2026-Present — Columbia Statistics Merit Scholarship

Work Experience

Quantum Device Engineer and Data Scientist, HRL Laboratories, Malibu, CA

11/2021 – 04/2026

- Designed and built custom electrical systems for experimental quantum dot semiconductor device testing
- Performed electrical tests of complicated quantum dot semiconductor devices and maintained testing systems and liquid helium cryostats
- Led improvements in the design and fabrication of quantum dot semiconductor devices through the analysis of large data sets and in-depth studies of their electrical properties [2]
- Aggregated and analyzed device testing data for insights on process and fabrication improvements
- Managed and developed SQL databases for the aggregation and analysis of electrical test data and analysis
- Developed software to monitor fabrication health, identify problematic processes, & improve device design
- Characterized qubit yield of quantum dot semiconductor devices to determine effective tests and fabrication processes
- Applied physics-based mathematical models to better understand device failure mechanisms

Research Assistant, Space Science Laboratory, Berkeley, CA

05/2019 – 07/2021

- Published lead-author paper detailing the novel application of deep learning algorithms for the automation of data collection from the NASA Stardust mission. [1]
- Implemented deep learning methods to automate data collection
- Built 2D and 3D convolutional neural networks able to identify signatures of interplanetary dust collected by the NASA Stardust mission

Job Related Skills

- Fluency in many programming related topics
    - Languages: Python, SQL, Java, Labview, Matlab, Linux OS
    - Python Libraries: Pandas, Numpy, Tensorflow, Matplotlib/Seaborn, Scipy
- Knowledge of data science and statistics methods and concepts
    - Data augmentation and preprocessing for machine learning preparation
    - Failure modes effects analysis, Cluster analysis, Pareto analysis, PCA, Data Visualization
- Experience with version control, git software, and collaborative programming

Publications

1. **Jaeger, L.**, et al. “Automatic detection of impact craters on Al foils from the Stardust interstellar dust collector using convolutional neural networks” *Meteoritics and Planetary Science.* (2021)
2. **Jaeger, L**, *contributing author*, in Members of the HRL Quantum Team and Collaborators. "A Digitally Controlled Silicon Quantum Processing Unit." *Nature* 655, 1154–1159 (2026).